Multiple Region Proposal Experts Network for Wide-Scale Remote Sensing Object Detection

Qifeng Lin, Haibin Huang, Daoye Zhu, Nuo Chen, Gang Fu, and Yuanlong Yu

***Abstract*— Faced with the wide-scale characteristics of objects in optical remote sensing images, the current object detection models are always unable to provide satisfactory detection capabilities for remote sensing tasks. To achieve better wide-scale coverage for various remote sensing regions of interest, this article introduces a multiprediction mechanism to build a novel region generation model, namely, a multiple region proposal experts network (MRPENet). Meanwhile, to achieve both region proposal coverage and receptive field coverage of wide-scale objects, we constructed a prior design of an anchor (PDA) module and an adaptive features compensation (AFC) module to achieve the coverage of wide-scale remote sensing objects. To better utilize the multiexpert characteristics of our model, we customized a new training sample allocation strategy, dynamic scale-assigned expert learning (DSAEL), to cultivate the ability of experts to deal with objects at various scales. To the best of our knowledge, this is the first time that a multiple region proposal network (RPN) mechanism has been used in the object detection of optical remote sensing images. Extensive experiments have shown the generality and effectiveness of our MRPENet. Without bells and whistles, MRPENet achieves a new state-of-the-art (SOTA) on standard benchmarks, i.e., DOTA-v1.0 [82.02% mean average precision (mAP)], HRSC2016 (98.16% mAP), and FAIR1M-v1.0 (48.80% mAP).**

***Index Terms*— Adaptive features compensation (AFC), dynamic scale-assigned expert learning (DSAEL), multiprediction mechanism, object detection, remote sensing, wide-scale coverage.**

I. Introduction

As an essential branch of computer vision, remote sensing object detection focuses on recognizing objects with extreme scale variations, such as tiny vehicles with scales less than ten pixels and harbors with scales exceeding 1000 pixels in optical remote sensing images [1], [2]. Unlike general object detection, remote sensing object detection confronts a long-standing and unresolved issue: the wide-scale variation of objects. The detection of objects with wide-scale variations poses a significant challenge to traditional object detection methods, diminishing the robustness of prior approaches.

Recently, there has been a surge in scale detection studies on remote sensing objects. There are a large number of medium-scale (like 32^2–512^2) object recognition models based on traditional detectors [3], [4], [5], [6], [7], [8], [9]. It also includes enhancing the sensitivity of the model to small objects (smaller than 32^2) by enhancing the expression of detailed features [10], [11], adjusting the loss function [12], [13], [14], utilizing an anchor-free method [15], or designing a pretrained backbone [16], [17], [18]. It is also possible to enhance the detection capabilities of large remote sensing objects (bigger than 512^2) by scaling the image [19], [20] or broadening the receptive field [21], [22], [23] and using multiview learning to mine missing objects [24]. There are also improvements aimed at addressing angular prediction misalignment [25]. However, it seems that no model can simultaneously achieve the recognition of wide-scale remote sensing objects (including small objects, medium objects, and large objects).

To achieve wide-scale object detection in optical remote sensing, the model needs to provide region proposals ranging from 16^2 to 1024^2. However, for a single model, accurately identifying both small and large objects still faces lots of challenges. The resolution for tiny objects is low, which makes them difficult to recognize. Additionally, huge objects often exceed the coverage of the anchor proposal. These limitations mean that the detection ability of large objects and small objects is always weak, especially for objects with dimensions smaller than 10 pixels or larger than 1000 pixels. As shown in Fig. 1(a), among various current detectors, their region proposal networks (RPNs) may potentially fail to return proposals that fall outside the minimum and maximum coverage range. Therefore, the existing region proposal methods [4] fail to ensure comprehensive RPN recall, which motivates us to depart from traditional network architectures.

Therefore, this article proposes MRPENet, utilizing the structural advantages of multiexpert, to achieve wide-scale detection in remote sensing scenes. To achieve coverage from small to large objects, we redesigned an RPN named multiple region proposal experts network (MRPENet). Our model enables different RPN experts to provide region proposals at distinct scales. As shown in Fig. 1(b), our approach involves training three region proposal experts with diverse

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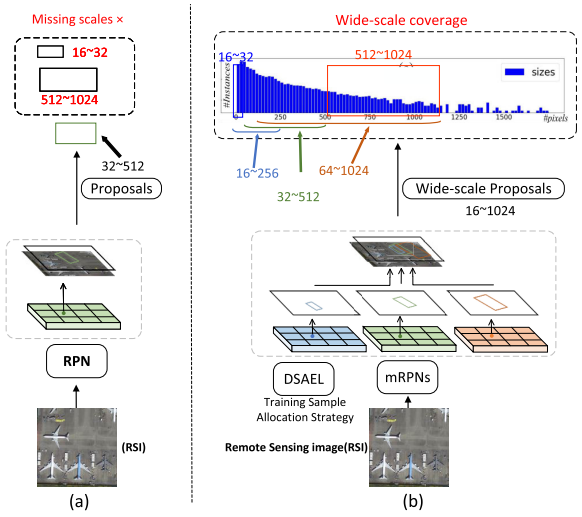


Fig. 1. (a) Existing detector, there is only one RPN to output a series of rectangular candidate proposals, which cannot cover the wide-scale remote sensing objects. (b) Proposed MRPENet leverages MRPEs to output wide-scale proposals for various scale object detection in optical remote sensing images (mRPNs: multiple RPNs).

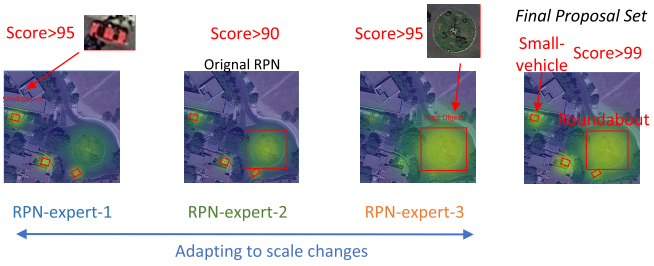


Fig. 2. Proposed MRPENet is a multiexpert model with three region proposal experts, each focusing on capturing remote sensing objects at different scales.

scale perceptions to cover objects across a wide range of scales effectively. Additionally, to address the challenge of obscure expert labeling, we customize dynamic scale-assigned expert learning (DSAEL). This learning strategy involves training multiple experts to handle samples with varying scales. As a result, each expert will perform well in specialized scale region recognition and outshine the others within its designated range. Simultaneously, we incorporate adaptive features compensation (AFC) for different RPN experts to adapt to variations in detection scales and enhanced semantic information at all scales, which involves anchor prior design and feature fusion. As shown in Fig. 2, MRPENet can adapt to scale variations of different remote sensing objects in the same image by multiple RPN experts, with expert-1 capturing smaller objects and expert-3 having better robustness to large objects.

The main contributions of this study are as follows.

1) An elegant MRPENet is proposed, which includes a multiple region proposal experts (MRPEs) module, a prior design of anchor (PDA) module, and an AFC module. The MRPE can achieve wide-scale coverage of the region to improve the recall rate of small and large objects. PDA and AFC involve the design of anchor priors and feature fusion, which are customized for different RPN experts to adapt to changes in detection scales.
2) A semi-supervised training strategy is customized for our detection framework, DSAEL, which can effectively improve the detection capabilities of our multiexpert model. At the same time, this strategy can automatically assign training labels to multiple region experts without requiring additional manual work to create expert labels, which is conducive to strengthening the promotion of our model.
3) The experimental results demonstrate the effectiveness and generality of our MRPENet for remote sensing objects. Our model achieved state-of-the-art (SOTA) detection accuracy, 82.02% [mean average precision (mAP)] on DOTA and 48.80% (mAP) on FAIR1M-1.0. To the best of our knowledge, this is the first time that a multiexpert mechanism has been applied to the wide-scale detection of remote sensing objects, which provides a new research idea to improve object detection performance in remote sensing.

The remainder of this article is organized as follows. Section II briefly reviews related work, including remote sensing image detection models, methods for handling scale variations, and multiexpert algorithms. Section III describes the overall architecture of the proposed model, in which we present the implementation of AFC for receptive field transformations on MRPENet, and the proposed semi-supervised DSAEL method, respectively. Section IV gives the related ablation study experiments, the experimental results of the three datasets, and the visualization analysis. Finally, Section V summarizes the article.

II. Related Work

A. Remote Sensing Object Detection

Most of the current mainstream high-performance remote sensing object detectors rely on the two-stage detector of the RCNN [4], which consists of a regional proposal network (RPN) and an RCNN detector head. In recent years, significant efforts have been made to enhance the backbone [22], [26], feature fusion in the neck module [27], refine the RPN [6], [28], [29], and develop specialized methods for extracting rotated regions of interest (ROI) [5]. Some other work focuses on improving regression losses [12], [13], [30], [31], [32] and label assignment [29], [33], [34], [35]. For example, GCL [14] optimizes the gradient of the rotated IoU loss to address the issues of angular error and scale sensitivity. ReDet [27] achieves this by constructing a rotation-equivariant ResNet and FPN [36] structure, enabling oriented RCNN [6] to generate precise proposals with rotation annotations. Gliding vertex [12] introduces four additional prediction variables for sliding vertices and a tilt factor to transform the horizontal proposals from the RPN into the final rotated objects. Oriented RepPoints [37] represent an object as a set of sample points. Recently, some works are highly significant, GeoKR [18] provide supervision for representation learning and network pretraining. AQE-Det [25] proposes an angle quality estimation method, which transforms angle prediction

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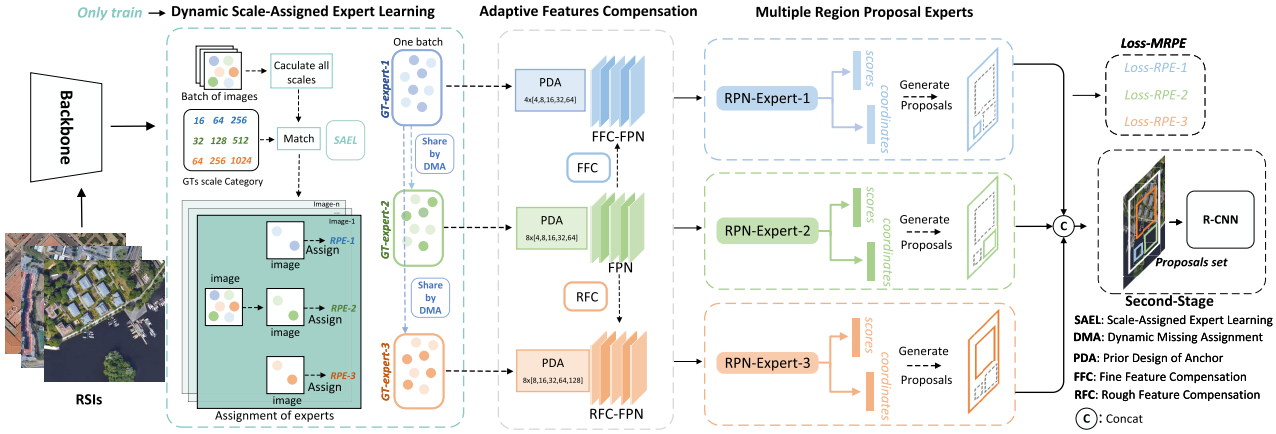


Fig. 3. MRPENet architecture: the framework consists of a backbone network, DSAEL module, AFC, three parallel RPN expert branches, and a second-stage detect head.

into a distribution estimation task, avoiding prediction errors caused by angle periodicity. MOL [24] reduces background interference through multi-instance learning combined with a temporal consistency strategy, and dynamically mines missing objects via multiview learning, significantly improving detection accuracy. In contrast with the above methods, MRPENet does not simply increase the number of anchors to improve the recall rate, it is also committed to providing higher quality proposals.

B. Methods for Handling Scale Variation

The scale variation is considered one of the most challenging problems in the multiscale object detection tasks [38], [39], [40], [41]. Small objects [42] lack rich detailed features, such as texture and color found in conventional-scaled objects, and models may easily misinterpret them as noise. Therefore, multilevel features are typically employed to mitigate scale variations and obtain high-resolution and high-semantic features. FPN [36] enhances the bottom layers' semantic information by passing high-level semantic features downward. SCRDet [43] adjusts the resolution of feature maps, selects appropriate scales, and uses simple feature fusion to ensure rich semantic and positional information. SNIP [44] proposes a scale-normalization method that selectively trains objects of appropriate scales at each image scale, resulting in objects within the desired scale range. TridentNet [21] designs multiple feature layer branches with different scales of object representation capabilities. Although these methods have shown promising results in solving multiscale challenges, there are still many difficulties in detecting extremely large and extremely small objects simultaneously. Therefore, we propose MRPENet to improve feature representation by compensating for features through multiple parallel branches.

C. Multi-Expert Algorithm

The core concept of the multiexpert mechanism [45], [46], [47], [48] is to divide training samples into subsets based on certain classification criteria. By leveraging the capabilities of each expert in their respective domains, the overall algorithm performance is significantly enhanced through collaborative efforts. Some efforts have been made to integrate the multiexpert concept into deep learning models, which aim to train multiple deep models and enhance recognition accuracy. For instance, EoDM [46] constructs two distinct models based on VGG16 [49] and MobileNet [50]. Additionally, some multi-expert methods, such as MoE [47], [51], [52], directly generate a weight distribution to unify the outputs of multiple experts, resulting in a single final prediction. MEDNet [10] constructs multiple feature pyramids (MFPs) to replace the traditional single feature pyramid to enrich the semantic representation ability of the model. For ME RCNN [45], based on input, only one expert assigned by the expert allocation network (EAN) can perform the detection task and generate the final prediction. However, this approach faces limitations in terms of the accuracy of weight distribution, which may impact the final prediction outcome. The multiexpert network designed in this article eliminates this decision-making process, and we use a semi-supervised DSAEL training strategy to allow experts to make independent predictions and improve object recall with MRPEs.

III. Method

This section first introduces the overall architecture of MRPEs designed to achieve wide-scale distribution detection. To achieve scale coverage for wide-scale remote sensing objects, we propose the PDA for region proposal coverage and the AFC for receptive field coverage. Furthermore, to train MRPEs without expert labels, we propose a DSAEL semi-supervised training strategy to further improve the detection accuracy of our MRPENet.

A. Overall Architecture

The overall architecture of our MRPENet is shown in Fig. 3. We introduced a module called MRPEs, which replaces the conventional RPN while adhering to the standard two-stage detection architecture. By implementing an AFC module, we constructed three feature pyramid networks with different receptive fields for different detection scales of three region proposal experts. The AFC module enables the customization

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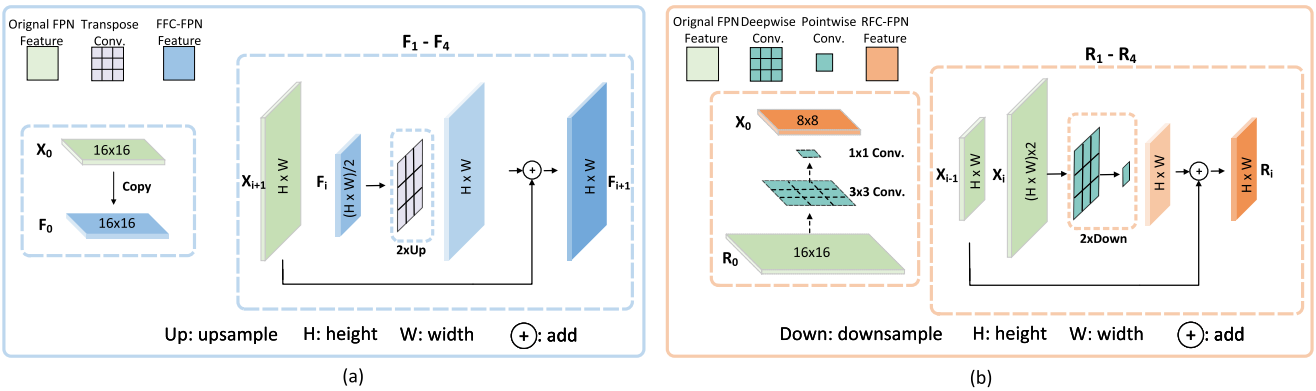


Fig. 4. AFC module is the reconstructed FPN (a) FFC for RPE-1 and (b) RFC for RPE-3.

of anchor prior configurations for each RPN expert, facilitating efficient object detection within their respective preferred scale ranges. Additionally, we designed a DSAEL strategy tailored specifically designed for training our MRPENet. The DSAEL strategy involves distributing actual annotation data among the three experts during the training phase to optimize their learning process.

In the proposed modules, the forward process begins by applying the DSAEL strategy to allocate training samples, dividing them into three sets based on scale differences, and then routing them through the corresponding RPN heads. To obtain better semantic representation, particularly in terms of receptive field coverage for both small and large objects, the AFC utilizes two modules: fine feature compensation (FFC) and rough feature compensation (RFC). These modules work together to construct two new feature pyramids, namely the FFC-FPN and RFC-FPN. The first expert handles the FFC-FPN output, which is enriched with fine-grained features via the FFC module. The third expert deals with the RFC-FPN output, augmented with additional semantic information through the RFC module. The second expert processes the standard FPN output from the backbone network. This division ensures that each expert operates within its range of feature representation expertise, thereby generating proposals of varied scales. These proposals are subsequently aggregated and fed into the RCNN for further processing.

By employing this approach, our architecture efficiently leverages the complementarity of features and scales to boost object detection accuracy. Our innovation extends beyond the implementation of a multiexpert mechanism; it encompasses how each expert utilizes tailored feature compensation and proposal generation mechanisms to detect objects across diverse scales. We are confident that this MRPE system, coupled with advanced training strategies, can substantially enhance the performance and adaptability of detection models.

B. Wide-Scale Objects Coverage

To detect wide-scale remote sensing objects, two prerequisites need to be guaranteed. First, to ensure that the detector has a high recall rate for wide-scale objects, the region proposals generated by the detector must be able to complete the proposal coverage of wide-scale objects. Second, to ensure that the feature map can achieve the feature representation capability of wide-scale objects, the detector must achieve receptive field coverage of the wide-scale region of interest. Therefore, in our MRPENet, the PDA is designed to achieve coverage of region proposals, and the AFC is customized to achieve receptive field coverage of wide-scale objects.

1) Prior Design of Anchor: To tackle the challenge observed in the current anchor-based method, where matching small objects with high-quality positive samples is difficult and large objects may not correspond to an anchor at all, we designed a novel anchor solution. This approach involves a fresh perspective on anchor configurations. To avoid affecting the recall rate of the original detection scales when expanding the detection range, we retain the original RPN detection structure. Then, based on receptive field theory, our model expands the detection scales for smaller objects and larger objects from two directions. In detail, for RPE-1, we maintain the same stride of baseline RPN but implement a strategy of decreasing the anchor scale; the anchor sizes are adjusted to $4 \times [4, 8, 16, 32, 64]$. Consequently, the Region Proposal Expert-1 (RPE-1) can cover small-scale objects with sizes ranging from 16 to 256. Region Proposal Expert-2 (RPE-2) follows the conventional RPN design, covering objects with scales ranging from 32 to 512. Region Proposal Expert-3 (RPE-3) adopts a constant anchor scale by modifying the stride of the feature map relative to the original map. This adjustment enables the expert to handle large objects using larger anchors mapped back to the original map, covering very large objects ranging in size from $8 \times [8, 16, 32, 64, 128]$, i.e., scales of 64 to 1,024. Similar to Faster RCNN [4], we employ anchors with multiple aspect ratios {1:2, 1:1, 2:1} at each level, resulting in a total of 15 anchors across one pyramid. This approach strikes a balance by maintaining the performance of the original RPN while enhancing the model's capability to detect both smaller and larger objects.

Since the high-performance detectors in the current field of remote sensing object detection are all two-stage detectors, the proposed MRPENet is also based on a two-stage architecture. Accordingly, we have devised a customized scale distribution

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for region proposals tailored to each expert. That is, each expert provides five scales of region proposals. For instance, RPE-1 provides region proposals with scales of (16, 32, 64, 128, 256), RPE-2 provides region proposals with scales of (32, 64, 128, 256, 512), and RPE-3 provides region proposals with scales of (64, 128, 256, 512, 1024).

Building upon this foundation, the scale predefinition is specifically aimed at detecting objects with a wide-scale distribution. Currently, almost all existing remote sensing datasets fall within this range. However, for datasets that extend beyond this range, the scale distributions of the RPEs can be fine-tuned using the "PDA" method to extend the detection capabilities further. This adaptability ensures that our anchor solution maintains its effectiveness across a wider range of object sizes, enhancing the model's suitability for various detection scenarios.

2) Adaptive Features Compensation: The PDA acts as a foundational prerequisite for AFC. We initially conceptualize the prior anchors, and then, leveraging PDA, we perform feature compensation across different FPN layers (i.e., X_4: 256 × 256, X_3: 128 × 128, X_2: 64 × 64, X_1: 32 × 32, and X_0: 16 × 16), highlighting a symbiotic relationship.

For RPE-1, we employ the FFC for the input feature layer. Since transposed convolution [53] can fill in missing details through interpolation, it increases the complexity and information of the feature map and helps the network learn more detailed features. To achieve this, we use a 3 × 3 transposed convolution with learnable weights for upsampling (with a factor of 2) to obtain high-resolution feature maps. Subsequently, we merge the upsampled feature maps with the original feature maps, ensuring that their dimensions are consistent for effective fusion. For an input size of 1024, our FFC-FPN incorporates five feature maps of varying sizes (i.e., F_4: 256 × 256, F_3: 128 × 128, F_2: 64 × 64, F_1: 32 × 32, and F_0: 16 × 16). The specific implementation is

$$\widetilde{\mathbf{X}} = [\mathbf{X}_0, \mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_4] \tag{1}$$

$$\mathbf{F}_0 = \mathbf{X}_0 \tag{2}$$

$$\mathbf{F}_{i+1} = \mathbf{X}_{i+1} + \mathcal{F}_{\mathbf{up}}^{3\times3}(\mathbf{F}_i), \quad \text{for} \quad i \text{ in } [0, 3] \tag{3}$$

where $\widetilde{\mathbf{X}}$ is the output of FPN, $\mathbf{X_i} \in \mathbb{R}^{C\times H\times W}$, $\mathcal{F}_{\mathbf{up}}^{3\times3}(\cdot)$ is the convolution kernel size of 3 × 3 transposed convolution. Fig. 4(a) shows the FFC processes.

For RPE-3, we employ the RFC for the input feature layer. Utilizing depth separable convolution [54] is beneficial as it reduces the number of model parameters and contributes to better learning of large receptive field features by convolving the features of each channel and then combining them. Initially, we use a depth separable convolution with channels of 256 to downsample the feature maps by two times, reducing them to half of the original size. Additionally, the original size of 16 × 16 is downsampled to an 8 × 8 feature layer. The remaining four feature layers, along with the original feature maps, maintain the same size across the FPN. Consequently, for an input size of 1024, our RFC-FPN comprises five feature maps of different sizes (i.e., R_4: 128 × 128, R_3: 64 × 64, R_2: 32 × 32, R_1: 16 × 16, and R_0: 8 × 8). The specific implementation is

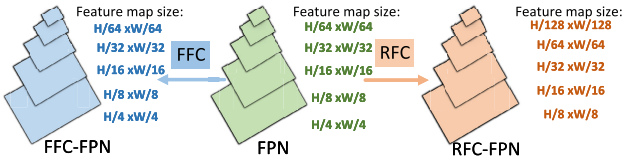


Fig. 5. FFC-FPN and RFC-FPN are two feature pyramids obtained from FPN by using FFC and RFC in the AFC module, respectively. FFC does not change the size of the feature map, thus compensating for more detailed features, while RFC changes the size of the feature map, thus compensating for semantic information. Here, H and W represent the height and width of the input image.

$$\mathbf{R}_i = \mathcal{F}_{\mathbf{pw}}^{1\times1}\left(\mathcal{F}_{\mathbf{dw}}^{3\times3}(\mathbf{X}_i)\right) + \mathbf{X}_{i-1}, \quad \text{for} \quad i \quad \text{in} \quad [1, 4] \tag{4}$$

$$\mathbf{R}_0 = \mathcal{F}_{\mathbf{pw}}^{1\times1}\left(\mathcal{F}_{\mathbf{dw}}^{3\times3}(\mathbf{X}_0)\right) \tag{5}$$

where $\widetilde{\mathbf{X}}$ is the output of FPN, $\mathbf{X_i} \in \mathbb{R}^{C\times H\times W}$, $\mathcal{F}_{\mathbf{pw}}^{1\times1}(\cdot)$ and $\mathcal{F}_{\mathbf{dw}}^{3\times3}(\cdot)$ are convolution kernels with sizes of 1 × 1 for pointwise convolution and 3 × 3 for depthwise convolution combinations. Fig. 4(b) shows the formation of R_0 and R_1 in the RFC module. It visualizes the process of RFC to expand the receptive field of the feature layer, which is more conducive to the detection of large objects by the RPE-3. It is worth noting that our AFC module adds only one transposed convolutional layer and one depth-separable convolution, and the layers of the features compensation pyramid share these convolutions with each other.

Leveraging various strides anchors from PDA and diverse feature layers of the FPNs generated by AFC, our MRPE can output a large number of RoI features to achieve wide-scale object detection. Specifically, FFC is meticulously designed for RPE-1 to enhance its sensitivity to the detailed features of small objects, while RFC is tailored for RPE-3 to improve its detection capabilities for larger objects. This approach ensures that RPE-1 is dedicated to detecting smaller objects, and RPE-3 is focused on enhancing the recognition of larger objects. Meanwhile, RPE-2 performs well in detecting an object of normal-scale objects. Fig. 5 shows in detail the feature map sizes of different FPNs after feature compensation.

C. Dynamic SAEL

1) Scale-Assigned Expert Learning (SAEL): Considering the fact that the three RPN expert branches would provide different ranges of object proposals, it is prudent to enable multiple experts to specialize in different scale samples independently. To train each RPN expert to excel in generating region proposals with various scale intervals, scale-assigned expert learning (SAEL) is introduced into our module.

Since the category labels and coordinate labels are already present in the labels of the dataset for object detection, we calculated the scale of the ground truth (GT) labels matched by each region proposal by

$$\mathbf{Scales(GT)} = \sqrt{w^2 + h^2} \tag{6}$$

where w is the width of the GT boxes and h is the height of the GT boxes, we use the scale of these GT boxes to assign them to different experts for expert loss learning.

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As described in Algorithm 1, based on the designs of both PDA and AFC, we first assign different detection scales to different RPEs. Specifically, we assign the scales of 16, 64, and 256 to RPE-1, the scales of 32, 128, and 512 to RPE-2, and the scales of 64, 256, and 1024 to RPE-3. Secondly, during each round of batch processing, the region proposals matched to the corresponding GT are assigned to the experts, which has the detection scale closest to the scale of this GT. Then, based on the losses between the assigned region proposals and the matched GTs, the loss learning is carried out by RPE-1, RPE-2, and RPE-3 simultaneously. Since greater diversity in scale distributions facilitates experts in acquiring varied knowledge, different scale distributions are assigned to different region proposal experts. The details are as follows:

$$L_{\mathbf{reg}} = \begin{cases} L_{\mathbf{RPE-1-reg}} \\ \quad \text{if} \quad \text{scale}(\mathbf{GT}) \in (16, 64, 256) \\ L_{\mathbf{RPE-2-reg}} \\ \quad \text{if} \quad \text{scale}(\mathbf{GT}) \in (32, 128, 512) \\ L_{\mathbf{RPE-3-reg}} \\ \quad \text{if} \quad \text{scale}(\mathbf{GT}) \in (64, 256, 1024) \end{cases} \tag{7}$$

where L_{reg} is the total gain boundary loss of MRPE, $L_{\text{RPE}-i-\text{reg}}(i = 1, 2, 3)$, are divided into the boundary losses of RPE-1, RPE-2, and RPE-3, respectively. In RPN, there is no focus on the specific category, but only on whether it is a foreground object of interest or not. Therefore, the foreground loss of MRPE is the same as that of RPNs, and the foreground losses of the related GTs are back-propagated to the corresponding experts, which completes the loss learning of the expert network to judge whether a region is foreground or not. Specifically

$$L_{\mathbf{fore}} = \begin{cases} L_{\mathbf{RPE-1-fore}} \\ \quad \text{if} \quad \text{scale}(\mathbf{GT}) \in (16, 64, 256) \\ L_{\mathbf{RPE-2-fore}} \\ \quad \text{if} \quad \text{scale}(\mathbf{GT}) \in (32, 128, 512) \\ L_{\mathbf{RPE-3-fore}} \\ \quad \text{if} \quad \text{scale}(\mathbf{GT}) \in (64, 256, 1024) \end{cases} \tag{8}$$

where L_{fore} is the total foreground loss of MRPE, $L_{\text{RPE}-i-\text{fore}}(i = 1, 2, 3)$ are the foreground losses for RPE-1, RPE-2, and RPE-3, respectively.

Based on the definitions of total boundary loss and total foreground loss for MRPE, MRPENet can perform loss learning based on the loss function to complete the training of the detection network

$$\text{Loss}_{\mathbf{MRPE}} = \frac{1}{N_{\text{fore}}} \sum_i L_{\text{fore}} + \frac{1}{N_{\text{reg}}} \sum_i P_i^* L_{\text{reg}} \tag{9}$$

where the label P^* is set to 1 when the ith region proposal mapped by the GT is in the foreground and 0 otherwise.

2) Dynamic Missing Assignment: During the implementation of the SAEL strategy discussed previously, there is a possibility that an expert may not have any matched samples during a batch process. This situation can easily result in region proposal experts without matched samples being unable to be effectively trained. To avoid ineffective training, we introduce a strategy of dynamic missing assignment (DMA). In cases where RPE-2 fails to match any samples within a batch, the samples from RPE-1 are shared with RPE-2 to ensure continuity of learning opportunities. Similarly, if RPE-3 finds itself without allocated samples, it will also receive samples from other RPEs. The assignment method of sample missing enhances the learning efficiency of our model.

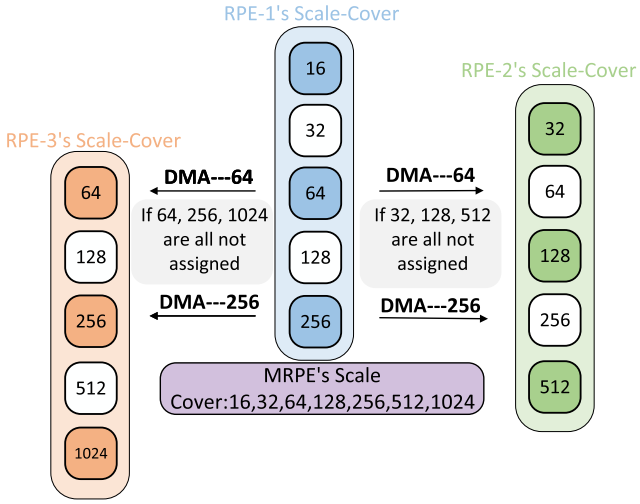


Fig. 6. Processes of DMA.

The underlying logic of this strategy is rooted in the scale adjacency of the experts and the partial scale overlap between RPE-1 and RPE-3, allowing for the sharing of RPE-1 samples with either RPE-2 or RPE-3. Fig. 1(b) demonstrates that in the remote sensing dataset, the objects within the range of 16–256 are significantly more abundant than those in the two scale ranges of 32–512 and 64–1024. Therefore, RPE-1 usually be assigned to more objects so that we can utilize RPE-1 to compensate for the other two experts. If RPE-2 or RPE-3 are not assigned samples, RPE-1 can share assigned samples with RPE-2 and RPE-3 based on the intersection of the scales from RPE-1 to RPE-2 and RPE-1 to RPE-3. As shown in Fig. 6, this strategy effectively addresses potential sample imbalances during each training iteration, ensuring that RPE-2 and RPE-3 have sufficient samples for robust training. By dynamically allocating the matched scale sample of RPE-1 to either RPE-2 or RPE-3, our DMA strategy effectively prevents ineffective training across all Region Proposal Experts.

Given these considerations, our training methodology is aptly named DSAEL, emphasizing its focus on DMA learning (i.e., DMA) and the selective allocation of training samples based on scale (i.e., SAEL). This strategy not only optimizes the training process but also significantly enhances the model's ability to learn from a wide range of object scales, ensuring comprehensive and effective training for all involved experts. In the application of multiexpert algorithms, the training of multiple experts typically relies on specialized multiexpert labels. However, for our MRPENet system, the GT in the training data consists only of category labels and boundary labels, without including labels specifically designed

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Algorithm 1 SAEL

Assign the scales 16, 64, 256 to RPE-1;
Assign the scales 32, 128, 512 to RPE-2;
Assign the scales 64, 256, 1024 to RPE-3;
Calculate the scale of the GT labels matched by each region proposal:
Scales(**GT**) $= \sqrt{w^2 + h^2}$ ▷ w and h are the width and height of the GT boxes
for each region proposal **do**
    Calculate the absolute difference of (**GT**) with 16, 32, 64, 128, 256, 512, 1024
    if scale (**GT**) is closest 16 **then**
        Assign to RPE-1;
    else if scale (**GT**) is closest 32 **then**
        Assign to RPE-2;
    else if scale (**GT**) is closest 64 **then**
        Assign to RPE-1 and RPE-3;
    else if scale (**GT**) is closest 128 **then**
        Assign to RPE-2;
    else if scale (**GT**) is closest 256 **then**
        Assign to RPE-1 and RPE-3;
    else if scale (**GT**) is closest 512 **then**
        Assign to RPE-2;
    else if scale (**GT**) is closest 1024 **then**
        Assign to RPE-3;
    end if
end for
Perform loss learning by RPE-1, RPE-2, RPE-3
Optimize the MRPENet based on the loss functions: $\mathbf{Loss_{MRPE}} = \frac{1}{N_{fore}} \sum L_{fore} + \frac{1}{N_{reg}} \sum P_i^* L_{reg}$

for multiple experts. Under these circumstances, our training approach can be considered a new form of semi-supervised learning.

IV. Experiments

In this section, we will first evaluate the generality of our MRPEs across different detection frameworks. Subsequently, we used Oriented RCNN as the baseline to conduct the comprehensive ablation study on MRPENet to verify the effectiveness of each component. Leveraging a mainstream detection framework, we compare our MRPENet with the SOTA detectors on some public benchmarks for comparative analysis. In addition, we further demonstrate the effectiveness and advancement of our model through visualizations based on different datasets. This includes visual demonstrations by three experts using Eigen-CAM.

A. Datasets

DOTA-v1.0 [2] is a large-scale rotated object detection dataset containing 2,806 photographs and 188,282 instances with oriented bounding box annotations. The dataset covers fifteen object classes: Plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). The image size of the DOTA dataset is extensive, ranging from 800 × 800 to 4000 × 4000 pixels. DOTA-v1.5 adds more instance labeling to v1.0 and adds the container crane (CC) object category.

HRSC2016 [55] is another widely used arbitrary-oriented object detection benchmark. It contains 1061 images with sizes ranging from 300 × 300 to 1500 × 900. Both the training set (436 images) and validation set (181 images) are used for training and the remaining for testing. For the evaluation metrics on the HRSC2016, we report the COCO [56] style mAP as well as the average precision under 0.5 and 0.75 thresholds (AP50 and AP75). In the data pre-processing procedure, we do not change the aspect ratios of images.

FAIR1M-v1.0 [57] is a recently published remote sensing dataset that consists of 15 266 high-resolution images, and more than 1 million instances. It contains five categories and 37 subcategories objects.

B. Implementation Details

In our experiments, we utilize three datasets: DOTA-v1.0 [2], HRSC2016 [55], and FAIR1M-v1.0 [57], to validate the effectiveness of our proposed MRPENet. The results of DOTA, HRSC2016, and FAIR1M were generated using the MMRotate [58] toolbox.

To ensure fairness, we employ the same dataset treatment as other mainstream methods. All models undergo training on the training and validation sets and are tested on the test set. Next, we use the AdamW optimizer [59]. For the DOTA dataset, we train for 12 epochs, and for HRSC2016, we train for 36 epochs. The initial learning rate is set to 0.0001 for DOTA and FAIRM1.0, and 0.00005 for HRSC2016, with the weights decayed to 0.05. In both the DOTA and the FAIRM1.0 datasets, we use 4 RTX3090 GPUs with a batch size of 4 for model training and 1 RTX3090 GPU with a batch size of 1 for HRSC2016. On the DOTA dataset and FAIR1Mv-1.0, the original image is cropped into a 1024 × 1024 patch. The cropping stride is set to 824, i.e., the pixel overlap of 200 between two adjacent patches. In multiscale training and testing, we first rescale the original image to three scales (0.5, 1.0, and 1.5), and then crop it into 1024 × 1024 patches with a stride of 524, and add data enhancement with random rotations only during multiscale training.

C. Effectiveness on Various Architectures

The experimental results on the DOTA dataset, achieved by adding multi-RPN experts into mainstream high-performance models and comparing them with their baselines, are shown in Table I. It is easy to see that our MRPE module consistently improves accuracy on all detection models, which shows the effectiveness and versatility of the proposed MRPE module in various rotating object detection models.

To further verify the effectiveness and versatility of the proposed MRPE, we also compared the above detectors with and without our MRPE on the HRSC2016 dataset. As shown in Table II, our MRPE achieves a significant improvement

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TABLE I

EXPERIMENTAL RESULTS FOR VARIOUS DETECTORS ON THE DOTA-V1.0 DATASET. THE EXPERIMENTAL RESULTS VALIDATE THE EFFECTIVENESS AND COMPATIBILITY OF OUR MODEL ON VARIOUS DETECTION FRAMEWORKS. MRPE DENOTES THE PROPOSED MRPES

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	**mAP**
RoI Trans. [5]	R-50	88.98	82.17	54.58	76.28	79.28	77.96	87.94	90.90	87.19	85.65	61.44	62.64	74.64	72.43	59.22	76.08
RoI Trans. + **MRPE**	R-50	88.70	**84.03**	53.76	72.15	79.75	**82.47**	**88.42**	90.85	**87.92**	**85.77**	**67.45**	**66.86**	**76.57**	**73.47**	**62.51**	**77.37** ↑(1.29)
Rotate Faster RCNN [4]	R-50	89.40	81.81	47.28	67.44	73.96	73.12	85.03	90.90	85.15	84.90	56.60	64.77	64.7	70.28	62.22	73.17
Rotate Faster RCNN + **MRPE**	R-50	89.01	**83.68**	**51.43**	**73.82**	**79.29**	**74.09**	**87.07**	90.86	**85.76**	84.88	**61.15**	62.02	**71.61**	**72.42**	59.66	**75.18** ↑(2.01)
Oriented RCNN [6]	R-50	89.35	81.41	52.71	75.03	79.04	82.41	87.82	90.89	86.40	85.29	63.36	65.70	68.28	70.48	57.23	75.69
Oriented RCNN + **MRPE**	R-50	88.99	**84.08**	**53.58**	**79.11**	**79.13**	**83.09**	**88.42**	90.86	86.25	84.51	**68.79**	**65.86**	**76.35**	**71.40**	**58.47**	**77.26** ↑(1.57)

TABLE II

EXPERIMENT RESULTS ON THE HRSC2016 DATASET. THE MRPENET IS ALSO EFFECTIVE ON THE HRSC2016 DATASET ON SOME POPULAR OBJECT DETECTORS

Method	AP50	AP75	mAP
RoI Trans. [5]	90.30	80.00	65.56
RoI Trans. + **MRPE**	**90.40**	**88.30**	**70.29** ↑(4.73)
Rotate Faster RCNN [4]	83.30	37.10	40.58
Rotate Faster RCNN + **MRPE**	**86.30**	**48.70**	**47.27** ↑(6.69)
Oriented RCNN [6]	90.40	81.10	68.80
Oriented RCNN + **MRPE**	**90.60**	**89.20**	**72.09** ↑(3.29)

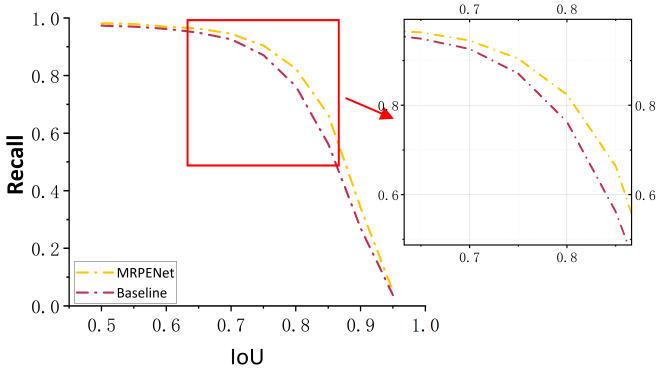


Fig. 7. Recall comparison between MRPENet and the baseline model, Oriented RCNN.

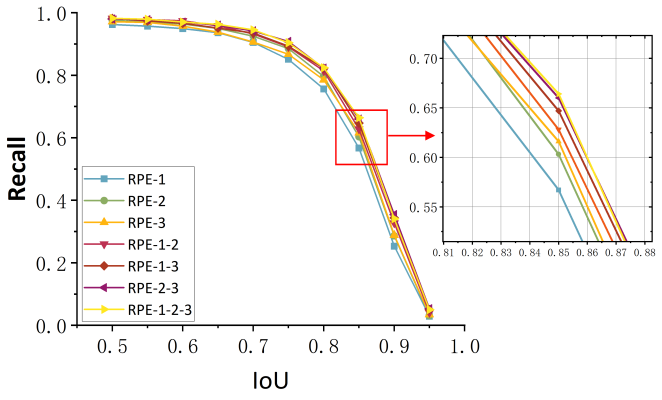


Fig. 8. Comparison of MRPENet's recall under different expert pairings.

of more than 4.73% mAP on the RoI Trans framework, and 6.69% and 3.69% mAP for Rotated Faster RCNN and Oriented RCNN, respectively.

The above experiments demonstrate that the proposed MRPE is compatible with various detection frameworks and

TABLE III

ABLATION INVESTIGATES THE IMPACT ON THE PERFORMANCE OF THE WIDE-SCALE OBJECTS COVERAGE MODULE. THESE EXPERIMENTS ARE BASED ON THE DOTA DATASET AND WERE CONDUCTED USING ORIENTED RCNN AS A BASELINE

Model	Change Anchor Basesize Only	**PDA**	**AFC**		mAP(%)
			FFC	RFC	
Baseline	-				75.69
MRPENet	✓	-			76.12
	-	✓	-	-	75.04
		✓	✓	-	75.51
		✓	-	✓	76.45
		✓	✓	✓	**77.26**

TABLE IV

ABLATION INVESTIGATES THE IMPACT OF SAEL WITH DMA ON THE NETWORK PERFORMANCE OF OUR PROPOSED MULTI-EXPERT NETWORK. THE EXPERIMENTS ARE CONDUCTED USING THE DOTA DATASET, WITH ORIENTED RCNN AS THE BASELINE

Model	**Dynamic Scale-Assigned Expert Learning**		mAP(%)
	SAEL	DMA	
Baseline	-	-	75.69
MRPENet	-	-	76.89
	✓	-	76.67
	✓	✓	**77.26**

effectively enhances the detection performance for wide-scale remote sensing objects. As shown in Fig. 7, the oriented RCNN is used as our baseline model, and it is easy to find that our MRPE helps the baseline detector achieve a significant improvement in recall across all IoU thresholds, which demonstrates the effectiveness of our MRPE. This highlights the ability of our proposed model to enhance overall detection performance by optimizing recall. In Fig. 8, as the IoU threshold increases, the model recall of our three expert combinations significantly outperforms other combinations. This demonstrates that our model excels in capturing more objects in the image under strict high IoU conditions, thereby improving the recall rates of the detector, which means our model's capability to achieve wide-scale proposal coverage, and then enhance overall detection performance.

D. Ablation Study

In Section IV-C, we have verified the benefits of our MRPE. Here, we will further verify how each component of the proposed MRPE mechanism works. To make a fair comparison,

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TABLE V

ABLATION INVESTIGATES THE IMPACT OF THE NUMBER OF PROPOSALS GENERATED BY EACH EXPERT IN MRPENET ON THE MODEL PREDICTIONS, WITH ORIENTED RCNN AS THE BASELINE

Propsals \ AP	BD	GTF	HA	SP	SBF	mAP(%)
500 1000 1500	81.53	76.00	76.10	69.34	66.47	76.51
1500 1000 500	82.26	77.83	75.47	71.16	65.87	76.91
1000 1000 1000	**84.07**	**79.11**	**76.35**	**71.40**	**68.79**	**77.26**
2000 2000 2000	82.40	78.55	75.36	69.96	66.50	76.78

TABLE VI

ABLATION INVESTIGATES THE IMPACT OF DIFFERENT RCNN SAMPLE NUMBERS WITH POSITIVE SAMPLE RATE ON THE MODEL PERFORMANCE

RCNN Sample Nums	Positive Sample Rate	mAP(%)
512	0.25	76.87
768	0.25	76.93
1024	0.25	77.18
1024	0.5	**77.26**

we use the same oriented RCNN detection framework as the baseline model for all our ablation studies and adopt the default hyperparameters defined in the MMRotate toolbox.

1) Evaluation on Wide-Scale Objects Coverage: As shown in Table III, initially, we tested the model's performance by solely increasing the anchor scale while keeping the feature map stride unchanged and without incorporating the AFC module. This approach led to a modest improvement of 0.43% (75.69% versus 76.12%) in model performance, revealing the limitations of adjusting the anchor scale alone. Subsequently, even with the introduction of PDA, performance improvements remained unsatisfactory without the integration of AFC modules. This highlights the necessity of combining PDA with AFC to unlock its full potential, as PDA aims to adjust anchors for proposal coverage of wide-scale, and AFC is used to achieve receptive field coverage (i.e., feature coverage) across these scales. Employing PDA in conjunction with our two AFC modules, FFC and RFC can significantly enhance the model performance. The FFC module contributed to a performance increase of 0.47% (75.04% versus 75.51%), while the RFC module led to a more notable improvement of 1.41% (75.04% versus 76.45%). This significantly boosted the model's capability to handle wide-scale objects, especially enhancing the abilities of RPE-1 and RPE-3 to detect objects of varying scales.

In summary, our proposed MRPE network, through the effective integration of PDA with the FFC and RFC modules, achieved a remarkable overall performance improvement of 1.57% (75.69% versus 77.26%). This series of experiments validated the effectiveness of our design, particularly in enhancing the model's ability to recognize objects across different scales, demonstrating the key role of the PDA and AFC modules and their significant contribution to improving model performance.

2) Evaluation on DSAEL: To evaluate the effectiveness of the training strategy customized for our MRPENet, we conducted three sets of experiments. As shown in Table IV, the first training method directly uses training samples to train the multiexpert model; the second training method only performs SAEL; the third method adopts SAEL and DMA to cooperate in joint training. It is easy to draw the following conclusions.

1) Compared with the baseline method, even with direct training, our MRPE can achieve a significant improvement in accuracy (i.e., 1.2%, 75.69% versus 76.89%).
2) When we simply leverage SAEL to assign diverse scale samples to different experts, although it is not as good as the accuracy of direct method 1 training, there is still a certain performance improvement compared to the baseline method.
3) When using DSAEL, including SAEL and DMA, our method can achieve the best accuracy improvement for our MRPENet.

In summary, the customized DSAEL strategy can indeed help our MRPE better detect remote sensing objects with wide-scale distribution.

3) Evaluation on Different Proposal's Nums: To investigate the effect of the number of proposals per expert on the performance of the model, we conducted a series of experiments. As shown in Table V, we discovered that the number of proposals per expert significantly influences the model's recall rate. After experimenting with various proposal ratios, we found an optimal balance of 1000 proposals per expert. This decision was informed by observing that a fixed number of proposals increased training time and resulted in redundant, duplicate proposals, adversely affecting the model's performance. Hence, we settled on allocating 1000 proposals for each expert, culminating in a high-quality set of 3000 proposals. This strategy effectively balances the need to minimize training time and avoid proposal redundancy, while still enhancing the model's ability to accurately detect objects across large, medium, and small scales. Our findings, substantiated by Table V, confirm that a well-considered mix of proposal numbers plays a crucial role in optimizing MRPENet's performance.

4) Different RCNN Sample Numbers With Positive Sample Rate: For object detection, the increase in the number of proposals can improve the detection accuracy, but it also increases the cost of the detection head. As shown in Table VI, conventional two-stage detectors typically sample 512 instances with a positive sample rate of 0.25, which is not suitable for MRPENet. To solve this problem, we tried different numbers of samples and positive sample rates and found that a larger number of samples can significantly improve the model's ability to learn data features and distribution more comprehensively. An enhanced sample pool provides richer categorical insights, empowering the detection head to better distinguish between categories and boosting the model's overall capacity. We tried to set the number of random samples to 1.5 times the original number of proposals, and the final experiment showed that two times (i.e., 1024) was the most appropriate number for our MRPENet.

E. Complexity Analysis of MRPENet

For experiments on the HRSC2016 dataset, the model input is (3, 800, 512), and for experiments on the DOTA-v1.0

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TABLE VII
COMPARISON OF THE COMPUTATIONAL COST, FPS, AND ACCURACY BETWEEN OUR MODEL AND THE BASELINE MODEL

Model	Dataset	Input Size	#P(M)	FLOPs[1]	FPS[2]	mAP(%)
Oriented RCNN [6]	DOTA	(3, 1024, 1024)	41.15	211.35G	**21.3**	75.69
Oriented RCNN + **MRPE**	DOTA	(3, 1024, 1024)	43.59	315.4G	13.1	**77.26** ↑1.57
Oriented RCNN [6]	HRSC2016	(3, 800, 512)	41.13	91.06G	**27.8**	68.80
Oriented-RCNN + **MRPE**	HRSC2016	(3, 800, 512)	43.58	131.68G	21.7	**72.09** ↑3.29

[1] FLOPs are calculated for the whole network.
[2] FPS of our tests using a single RTX3090.

TABLE VIII
PERFORMANCE COMPARISON IN GENERATING VARYING NUMBERS OF PROPOSALS DURING THE TESTING PHASE

Model	Total Test Props.	FPS	mAP(%)
Oriented RCNN [6]	3000	16.3	**76.01**
Oriented RCNN [6]	2000	**21.3**	75.69
Oriented RCNN + **MRPE**	3000(1000 per RPE)	13.1	**77.26**
Oriented RCNN + **MRPE**	1800(600 per RPE)	**14.9**	77.22

TABLE IX
COMPARISON OF TRAINING COSTS BETWEEN OUR MRPENET AND THE BASELINE ORIENTED RCNN

Model	Dataset	Batch Size	TrainCost(hh:mm)
Oriented RCNN [6]	DOTA	2	03:33
Oriented RCNN + **MRPE**	DOTA	2	07:15 ↑03:42
Oriented RCNN [6]	HRSC2016	1	01:42
Oriented RCNN + **MRPE**	HRSC2016	1	02:09 ↑00:27

TABLE X
PERFORMANCE COMPARISON OF DIFFERENT MULTI-EXPERT MODELS ON DOTA-V1.5 DATASET, ALL MODEL USED FASTER RCNN AS THE BASELINE MODEL, INCLUDING MOE, ME-RCNN, MEDNET, AND OUR MRPENET

Method	Backbone	mAP	FPS
MoE [52]	R-50	65.34	8.11
ME-RCNN [45]	R-50	66.12	10.06
MEDNet [10]	R-50	67.36	10.32
MRPENet	R-50	**68.84**	**13.96**

dataset, the model input is (3, 1024, 1024). Overall, as shown in Table VII, for experiments on the HRSC2016 and DOTA dataset, our model only introduced a small number of parameters, but indeed increased the computational cost, and there was a decrease in frames per second (FPS). However, we achieved a more satisfactory improvement in accuracy.

Our work primarily focuses on improving the recall rate of the model. Therefore, the candidate proposals generated by multiple RPN experts significantly impact the final performance of the model. As evident in the first and second rows of Table VIII, for the baseline model, generating more proposals during testing increases the model's computational cost, and then the FPS will decrease. The third and fourth rows of Table VIII further validate this observation. However, compared to the baseline model, reducing the number of proposals during the test phase has less impact on the accuracy of our model. This suggests that as long as we provide enough proposals to the RCNN during the training phase, we can use fewer proposals during the testing phase to improve the FPS and achieve an increase in accuracy. Since our model incorporates multiple experts, there is a decrease in FPS even when reducing the number of proposals generated during testing. This is indeed an impact on our model, but the MRPENet can achieve considerable improvements in accuracy.

Since we added two additional RPNs in a two-stage approach to the currently dominant high-precision detector based on RPNs, the anchors generated by these RPNs require significant computational resources. As a result, the computational cost must be balanced against model performance. As shown in Table IX, for the same batch size, additional training time is added compared to the baseline, but this overhead is kept within acceptable limits to improve performance. As shown in Table VIII, there is no significant decrease in inference speed in real-world applications, making our model well-suited for detection scenarios requiring high accuracy.

F. Comparison With Detectors of Multi-Expert Algorithm

We compare MRPENet with other multiexpert CNN models in Table X. This experiment focuses on assessing the object detection performance of various multiexpert models on the DOTA-v1.5 dataset. To ensure fairness, all multiexpert models utilize three experts for performance comparison and use Faster RCNN as the baseline model. In addition to training three experts, MoE incorporates a gating network (GN) to adjust the weights of different experts. For ME RCNN, a four-step alternating algorithm is employed to train the three experts along with the expert assignment network (EAN). In contrast, MEDNet assigns training samples to different detection experts in an unsupervised manner using the loss distance k-expert clustering (LD-kEC) strategy.

It can be seen that our model's accuracy performance is very advantageous. MoE's experts selected through weights have a great deal of uncertainty, whereas ME RCNN's EAN outputs in the form of probabilities by assigning values to individual experts, as with MoE. Whereas our model is similar to MEDNet in that the model directly selects the best prediction from multiple outputs, which is obviously more conducive to obtaining higher detection accuracy, the DSAEL used by MRPENet is a semi-supervised training strategy, and compared to unsupervised training, semi-supervision is able to reduce the uncertainty in unsupervised through the guidance of labeled data, making the model more stable and reliable, and helping the model to better learn and distinguish between different classes or features, thus improving the performance of the model. In the tests, MoE, ME-RCNN,

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TABLE XI

COMPARISON WITH SOTA METHODS ON THE DOTA-V1.0 DATASET. THE ITEMS WITH RED AND BLUE COLORS INDICATE THE BEST AND SECOND-BEST RESULTS OF EACH COLUMN

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
One-stage																	
R3Det [11]	R-101	88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	86.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
DRN [60]	H-104	88.91	80.22	43.56	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
RSDet [13]	R-101	89.80	82.90	48.60	65.20	69.50	70.10	70.20	90.50	85.60	83.40	62.50	63.90	65.60	67.20	68.00	72.20
DAL [33]	R-50	88.68	76.55	45.08	66.80	67.00	76.76	79.74	90.84	79.54	78.45	57.71	62.27	69.05	73.14	60.11	71.44
G-Rep [61]	R-101	88.89	74.62	43.92	70.24	67.26	67.26	79.80	90.87	84.48	78.47	54.59	62.60	66.67	67.98	52.16	70.59
SASM [62]	RX-101	89.54	85.95	57.73	78.41	79.78	84.19	89.25	90.87	58.80	87.27	63.82	67.81	78.67	79.35	69.37	79.17
S^2ANet [8]	R-50	89.30	80.11	50.97	73.91	78.59	77.34	86.38	90.91	85.14	84.84	60.45	66.94	66.78	68.55	51.65	74.13
CFA [63]	R-50	89.08	83.20	54.37	66.87	81.23	90.96	87.17	90.21	84.32	86.09	52.34	69.94	75.52	80.76	67.96	76.67
Two-stage																	
RoI transformer [5]	R-101	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
SCRDet [43]	R-101	88.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
CSL [64]	R-50	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.1	68.93	76.17
CenterMap [65]	R-101	89.83	84.41	54.6	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06	76.03
FAOD [66]	R-101	90.21	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
G.V. [12]	R-50	89.64	85.00	52.26	77.34	73.01	73.14	86.85	90.74	79.02	86.81	59.55	70.91	72.94	70.89	57.32	75.02
ReDet [27]	Re-R50	88.81	82.48	60.83	80.82	78.34	86.06	88.31	90.87	88.77	87.03	68.65	66.90	79.26	79.71	74.67	80.10
AOPG [15]	R-50	89.88	85.57	60.90	81.51	78.70	85.29	88.85	90.89	87.60	87.65	71.66	68.69	82.31	77.32	73.10	80.66
KFIoU [32]	Swin-tiny	89.44	84.41	62.22	82.51	81.10	86.07	88.68	90.90	87.30	88.38	72.80	71.95	78.96	74.95	75.27	80.93
Oriented RCNN [6]	R-50	89.84	85.43	61.09	79.82	79.71	85.35	88.82	90.88	86.68	87.73	72.21	70.80	82.42	78.18	74.11	80.87
RVSA [17]	ViTAE-B	88.97	85.76	61.46	81.27	79.98	85.31	88.30	90.84	85.06	87.50	66.77	73.11	84.75	81.88	77.58	81.24
LSKNet-S [22]	LSKNet-S	89.57	86.34	63.13	86.67	82.20	86.10	88.66	90.89	88.41	87.42	71.72	69.58	78.88	81.77	76.52	81.64
AQE-Det + RoI-trans. [25]	R-50	89.23	85.35	60.26	80.13	80.27	85.58	88.46	90.88	87.51	88.07	69.80	70.32	80.97	82.16	74.03	80.87
GCL + RoI-trans. [14]	Swin-tiny	88.99	85.51	62.52	82.45	78.89	85.93	88.31	90.84	86.13	87.39	73.09	69.86	84.23	81.54	78.63	81.69
MRPENet + RoI-trans.	R-50	89.19	85.87	60.20	81.44	80.13	85.97	88.82	90.86	85.80	88.40	72.21	68.81	79.61	82.80	71.23	80.76
MRPENet + Oriented RCNN	R-50	89.32	85.08	61.49	81.56	79.61	86.05	88.85	90.87	85.64	87.97	73.70	71.62	82.66	81.07	72.39	81.19
MRPENet + RoI-trans.	Swin-tiny	89.39	85.82	62.48	82.81	81.30	86.39	88.97	90.87	88.02	88.76	77.02	69.93	79.33	81.99	77.21	82.02

TABLE XII

COMPARISONS OF FINE-GRAINED CATEGORY RESULTS WITH SOTA METHODS ON THE FAIR1M-V1.0 DATASET. * INDICATES THAT THE RESULTS REFER TO THE FAIR1M PAPER [57]

Coarse Category	Sub Category	Gliding Vertex [12]*	RetinaNet [67]*	Cascade RCNN [68]*	Faster RCNN [4]*	ROI Trans [5]*	Oriented RCNN [6]*	LSKNet-S [22]	**MRPENet**
Airplane	Boeing737	35.43	38.46	40.42	36.43	39.58	42.84	39.84	39.88
	Boeing747	47.88	55.36	52.86	50.68	73.56	87.61	84.97	86.48
	Boeing777	15.67	24.75	29.07	22.5	18.32	18.83	24.21	23.67
	Boeing787	48.32	51.81	52.47	51.86	56.43	62.92	56.48	56.95
	C919	0.01	0.81	0	0.01	0	22.17	24.17	23.70
	A220	40.11	40.5	44.37	47.81	47.67	47.87	52.2	51.27
	A321	39.31	41.06	38.35	43.83	49.91	70.25	73.31	72.09
	A330	16.54	18.02	26.55	17.66	27.64	73.34	72.82	75.79
	A350	16.56	19.94	17.54	19.95	31.79	77.19	75.83	78.40
	ARJ21	0.01	1.7	0	0.13	0	32.49	46.39	38.22
Ship	Passenger Ship	9.12	9.57	12.1	9.81	14.31	20.21	20.43	22.84
	Motorboat	23.34	22.55	28.84	28.78	28.07	72.13	71.38	72.05
	Fishing Boat	1.23	1.33	0.71	1.77	1.03	13.53	15.81	15.59
	Tugboat	15.67	16.37	15.35	17.65	14.32	35.5	32.84	38.75
	Engineering Ship	15.43	19.11	18.53	16.47	15.97	16.23	14.79	14.67
	Liquid Cargo Ship	15.32	14.26	14.63	16.19	18.04	26.49	25.37	28.35
	Dry Cargo Ship	25.43	24.7	25.15	27.06	26.02	38.43	41.29	40.47
	Warship	13.56	15.37	14.53	13.16	12.97	34.74	36.2	41.06
Vehicle	Small Car	66.23	65.2	68.19	68.42	68.8	74.25	76.34	75.87
	Bus	23.43	22.42	28.25	28.37	37.41	47.02	55.54	56.61
	Cargo Truck	46.78	44.17	48.62	51.24	53.96	50.22	55.84	56.24
	Dump Truck	36.56	35.37	40.4	43.6	45.68	57.56	61.57	61.64
	Van	53.78	52.44	58	57.51	58.39	75.22	76.71	77.31
	Trailer	14.32	19.17	13.66	15.03	16.22	20.91	21.46	22.97
	Tractor	16.39	1.28	0.91	3.04	5.13	2.99	7.19	8.61
	Excavator	16.92	17.03	16.45	17.99	22.17	19.95	25.73	29.15
	Truck Tractor	28.91	28.98	30.27	29.36	46.71	1.77	4.74	8.02
Court	Basketball Court	48.41	50.58	38.81	58.26	54.84	55.35	61.78	60.35
	Tennis Court	80.31	81.09	80.29	82.67	80.35	82.96	81.06	87.26
	Football Field	53.46	52.5	48.21	54.5	56.68	64.62	70.39	71.01
	Baseball Field	66.93	66.76	67.9	71.71	69.07	90.36	89.94	89.23
Road	Intersection	59.41	60.13	55.67	59.86	58.44	60.82	62.9	66.36
	Roundabout	16.25	17.41	20.35	16.92	18.58	20.47	27.00	27.00
	Bridge	10.39	12.58	12.62	11.87	31.81	33.4	39.51	41.41
mAP(%)		29.92	30.67	31.18	32.12	35.29	45.6	47.87	48.80

and MEDNet need to perform additional forward inference operations for GN, EAN, and two additional ROIs, which is more time-consuming compared to adding our RPN experts. Therefore, our MRPENet achieves better performance in both inference speed and detection accuracy.

G. Comparison With the SOTA Methods

1) Results on the DOTA-v1.0 Dataset: We compare the proposed methods with some recently proposed and so far the most advanced methods [14], [17], [22], [25], and the results are presented in Table XI, the MRPENet obtains 82.02% mAP which is very competitive with the current SOTA methods. In some challenging tiny objects such as SV, LV, SH, and huge objects like GTF, SF, and SBF, we rank in the top two in detection performance across all categories. It is evident that our models excel in the majority of categories. To show that our model outperforms the previous SOTA, we compare the latest SOTA model LSKNet. As shown in Fig. 9(a), our model performs better at detecting HA categories, avoiding the compartmentalization produced by the SOTA model for HA. In (b), MRPENet also performs better at detecting more

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TABLE XIII
COMPARISON WITH SOTA DETECTION METHODS ON THE HRSC2016 DATASET

Method	Backbone	mAP(07)	mAP(12)
RoI Trans [5]	R-101	86.20	-
G.V. [12]	R-50	88.20	-
R3Det [11]	R-101	89.26	96.01
DAL [33]	R-101	89.77	-
S2ANet [8]	R-101	90.17	95.01
ReDet [27]	ReR-50	90.46	97.63
O-RCNN [6]	R-50	90.40	96.50
LSKNet [22]	LSKNet-S	90.65	98.46
MRPENet	R-50	90.60	98.20

LVs, greatly reducing the probability of missed detection. Additionally, in (c) and (d), MRPENet also performs better at object localization, with a higher rate of correct classification.

2) Results on the FAIR1M-v1.0 Dataset: We also compare our MRPENet with seven other models on the FAIR1M-v1.0 dataset. As Table XII shows, by adding our MRPE into the LSKNet-S model, our MRPENet performs extremely well, and it achieves the best mAP, achieving SOTA mAP of 48.80%.

3) Results on the HRSC2016 Dataset: We evaluated the performance of MRPENet using oriented RCNN as a baseline against eight SOTA methods on the HRSC2016 dataset, and the results in Table XIII show that our MRPENet outperforms all the other methods, with the PASCAL VOC 2007 [69] and VOC 2012 [70] metrics obtained the second-best performance with mAP of 90.60% and 98.20%, respectively.

H. Visualization of Dataset

1) DOTA-v1.0 Test Result Visualization: For the DOTA-v1.0 dataset, our proposed MRPENet demonstrates a lower leakage rate in object detection across a wide range of scales. As shown in Fig. 10, we compare the MRPENet detection results with the oriented RCNN (baseline) on the test set. Our MRPENet is able to detect dense small objects, such as the SV in Fig. 10(c). The reason is that our model improves the quality of proposals via the tailored design of RPE-1 for tiny object detection so that small objects can be detected and localized well. For larger objects, our MRPE is especially suitable for objects with large horizontal and vertical dimensions, as shown in Fig. 10(a), (b), and (d). Here, the MRPENet demonstrates higher accuracy in both object detection and localization compared to the baseline model, without being affected by truncation issues. This advantage is primarily due to RPE-3's ability to generate large-scale proposals, effectively covering objects such as harbors and bridges. It addresses the issue of large objects being prone to multiple individual detection results [like the harbor in Fig. 10(b)]. This is due to the fact that current two-stage RCNN detectors are unable to generate large enough proposals to detect such large objects. This further emphasizes the advanced nature of our MRPENet.

As shown in Fig. 11, the detection results of MRPENet and LSKNet on the trainval set were compared with the GT of the trainval set. Compared to LSKNet's ability to identify unlabeled instances in the trainval set, our approach demonstrates strong generalization, which also demonstrates the advanced nature of our approach.

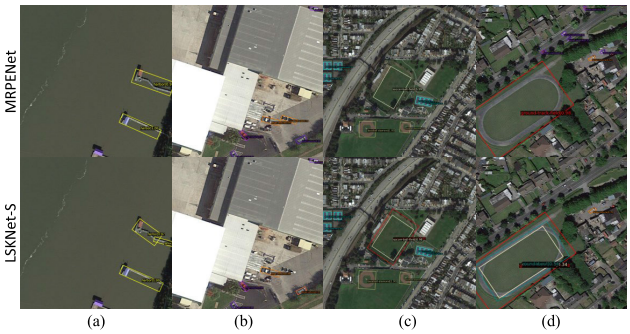


Fig. 9. Comparison of MRPENet + Oriented RCNN test results to baseline (Oriented RCNN) on the DOTA-v1.0 dataset.

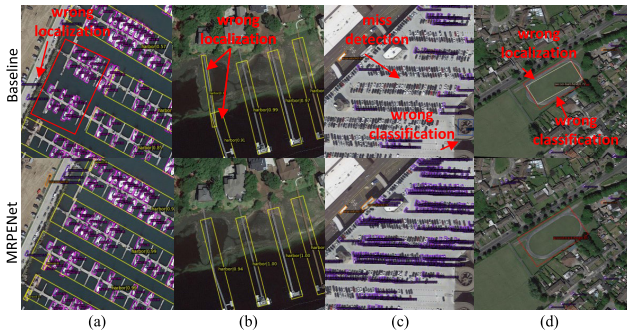


Fig. 10. Comparison of MRPENet + RoI Trans. (our best) test results to LSKNet (SOTA) on the DOTA-v1.0 dataset.

2) FAIR1M-1.0 Test Result Visualization: For the FAIR1M-v1.0 dataset, some results of our proposed model on the FAIR1M-v1.0 test set are also given in Fig. 12.

I. Visualization of Proposals

The MRPE generated a series of proposals, with the top 50 showcased in each image. As shown in Fig. 13, these regional proposal experts are capable of effectively identifying objects of various scales. Among the top 50 proposals, RPE-1 shows a significant overlap of proposals for small objects, whereas RPE-3 exhibits a high concentration in localizing large objects. Each of our experts demonstrates exceptional performance in detection tasks across various scales. Ultimately, by integrating the proposals of these three experts and then selecting the top 50, our MRPE achieves significant improvements in both density and localization compared to the baseline RPN.

J. Visualization of Eigen-CAM

Eigen-CAM employs principal component analysis (PCA) to produce heatmaps. Unlike depending on gradient information, Eigen-CAM extracts the main patterns of variation by conducting eigenvalue decomposition on the feature maps. This enables Eigen-CAM to capture the commonalities among different channels and generate a composite response map. We utilized ResNet-50 as the model's backbone to generate CAM heatmaps after passing through different RPEs.

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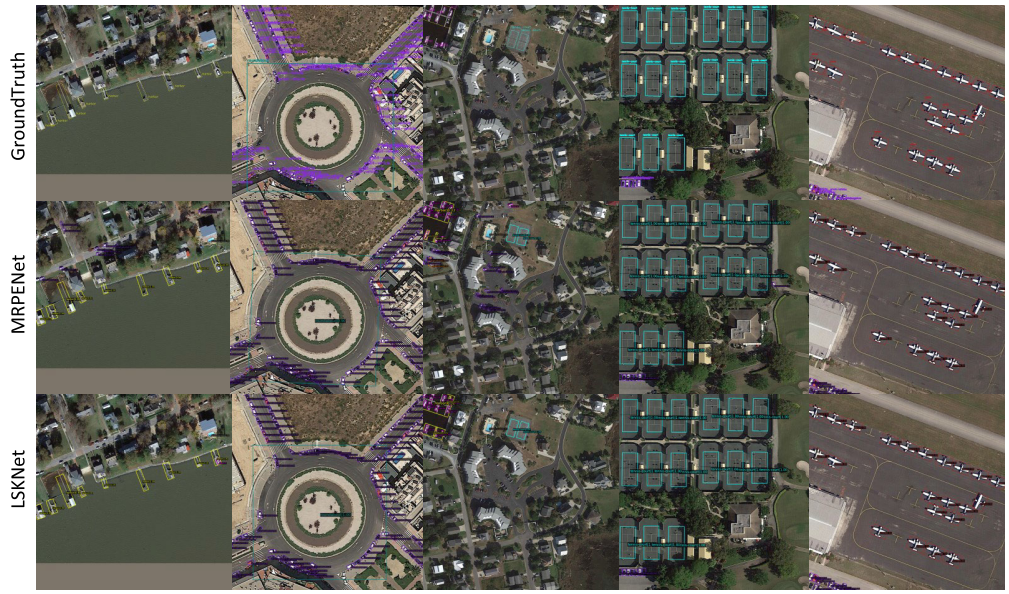


Fig. 11. Comparison of MRPENet + Oriented RCNN test results on DOTA-v1.0 trainval dataset. The first row contains the Ground Truth of the trainval dataset, and the second and third rows contain MRPENet and LSKNet detection results on the trainval dataset.

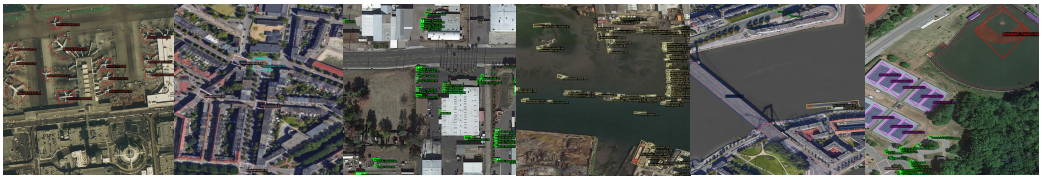


Fig. 12. Detection results of FAIR1M-v1.0 dataset from our MRPENet.

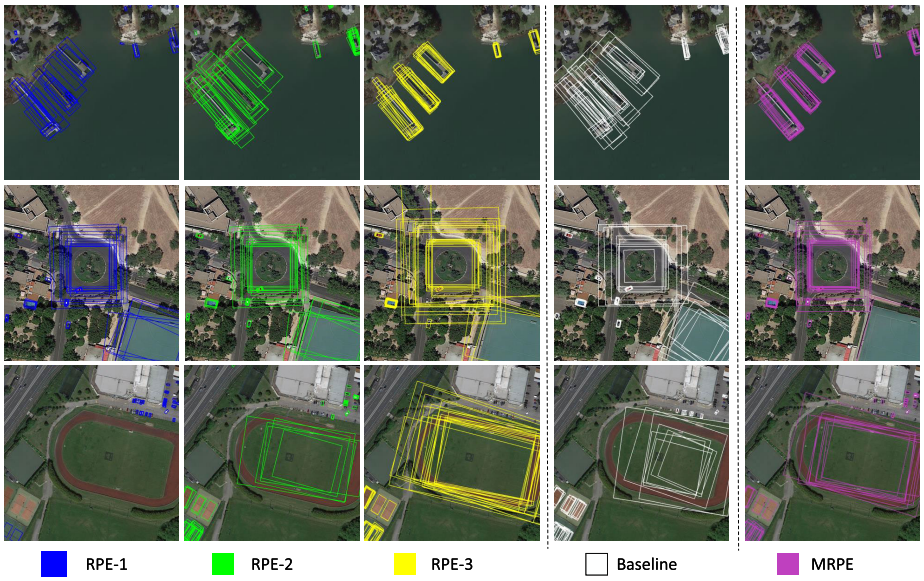


Fig. 13. Proposals were generated by MRPEs on the DOTA-v1.0 dataset. The first three columns display the proposals generated by each of the three experts, the fourth column shows the baseline RPN and the last column presents the top 50 of the proposals generated by our MRPE.

The visualization of Eigen-CAM is presented in Fig. 14, which depicts the value of each feature map's contribution to the final classification result. The figure reveals that due to the distinct scales covered by different experts, the scales of the categories vary as a result of the experts' diverse expertise. We display the CAM visualization of the 3×3 convolutional kernels from the three RPN expert networks. This clearly shows that the three experts learned different weight distributions for the same image during training. It can be observed that the attention diminishes from RPE-1 to RPE-3 for smaller objects, whereas it increases for larger objects and provides a clearer outline of the larger ones. For RPE-1, it exhibits

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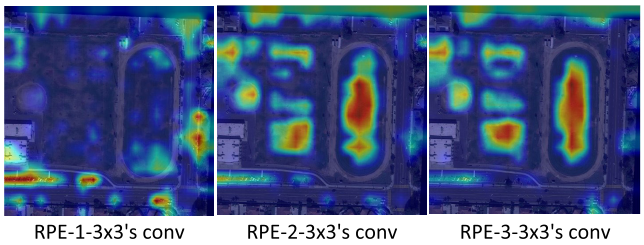


Fig. 14. Eigen-CAM visualization of the three Region Proposal Experts (RPEs).

a strong response to small-scale objects. In contrast, RPE-3 shows a strong response to large-scale objects with activation areas covering larger contiguous regions. This indicates that MRPENet indeed enables different RPNs to learn remote sensing features at various scales, thereby enhancing the model's ability to represent features of remote sensing objects at multiple scales.

V. Conclusion

In this article, we propose an MRPENet, designed for wide-scale object detection in remote sensing. Our model leverages three region proposal experts to improve the model's recall rate for various scale objects, thereby improving the model's detection accuracy. In our model, prior design anchor (PDA) and AFC are used to enhance the range of region proposals and feature representation capabilities for various scale objects. Then, a customized DSAEL is used to automatically assign different training samples to different experts, thereby strengthening the recall capabilities of experts for objects of varying scales. The extensive experiments demonstrate that the proposed MRPE-based detector consistently achieves SOTA performance on competitive remote sensing benchmarks, highlighting its effectiveness and versatility in remote sensing object detection tasks with various scale features.

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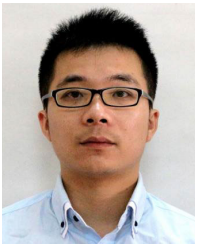
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